Exact Factorial Coordinates, Mark Costs, and Transport Bounds for Prime Predecessors

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M022

October 8, 2026

Abstract

We examine the finite-coordinate and probabilistic interfaces of *The Poisson–Dirichlet law for prime predecessors*. The arithmetic determinant, correlation, and long-prime estimates are identified as imported inputs, not independently established here. We give exact joint factorial Gram formulas, collapse the primewise memory identity by projective-line classes, and isolate a smaller sufficient memory cutoff for the source's absolute truncation bound. Exact Bernoulli-model mark moments expose the cost of removing rare marks and yield covariance-weighted alternatives under an explicit arithmetic comparison premise. A quantitative finite-prefix transport bound converts interior factorial-moment errors and missing mass into an ℓ^1-Wasserstein bound for ranked factors. Its qualitative consequence is a classical mass-conservation deduction, not a new prime-factor limit law. Entropy and sublinear power sums remain outside that conclusion without uniform integrability. Contribution assessment: C3/5, because the report proves source-specific conditional cost and transport refinements but supplies no new global arithmetic estimate.

1 Status, normalization, and scope

Applicability is the catalogue's **L3**; contribution is assessed separately as **C3/5**. "Source" below means the September 24, 2026 public manuscript [1]; "local" means a proof supplied in this report; "conditional" means that the stated arithmetic interface is assumed. No statement here certifies the whole source proof or the private corpus. The local algebra and measure theory do not prove the conjecture of Ford–Konyagin–Luca independently.

Let p be uniform among primes $3 \le p \le x$, with x real and $x \to \infty$. List the prime factors $q_i(p)$ of $p-1$ in decreasing order *with multiplicity*, padding with 1, and set

$$V_x = (V_{x,i})_{i\ge 1}, \qquad V_{x,i} = \frac{\log q_i(p)}{\log(p-1)}.$$

Thus V_x lies in the ranked mass-one space

$$\nabla_1 = \{v \in \ell^1 : v_1 \ge v_2 \ge \cdots \ge 0,\ \sum_i v_i = 1\}.$$

Write $\nabla_{\le 1}$ for the same space with sum at most 1. These spaces carry the ℓ^1 metric unless otherwise specified. Let U_i be independent uniform variables on $(0,1)$ and define

$$B_j = (1-U_j)\prod_{i<j} U_i, \qquad P = \mathrm{sort}(B_1, B_2, \ldots).$$

Then $\mathcal{L}(P) = \mathrm{PD}(1)$ and $\sum B_j = 1$ almost surely. The source claims convergence of every finite joint distribution of V_x to that of P. Its proof obtains size-biased no-escape before sorting. Our quantitative transport estimate uses precisely that interface.

Literal reading. The assigned source has 13 TeX files and 5,514 lines. We read 3,012 distinct TeX lines in bounded, untruncated displays: eight complete files and five partial files. The 291-line bibliography was read in full. Three applicable private originals were read over 382 distinct lines, with no claim to have read them in full. Exact inclusive intervals, pre-read SHA-256 values, final hashes, and complements are recorded in the owned read-scope and identity records. Structure-only searches earn no additional literal-read credit. The source README and the companion arithmetic manuscript are not scientific reading inputs to this report.

Provenance versus proof. The parent reports that the 15 source texts agree between captured commits `adc7f1241b42` and `fd4aeeb2ee4f`. Our own hashes identify the local files read. That agreement says nothing about correctness, current live-branch status, or external expert acceptance. Parent finite checks are not counted as author tests, and mutable parent records are not frozen by this report.

2 Whole-source dependency map

Use $L = \log x$. Fixed parameters are chosen before $x \to \infty$. All estimates allowing constants depending on a fixed number of mark groups must preserve that quantifier. The following map covers every scientific source section, but does not purport to check every proof in it.

Section	Role in the source	Inspection boundary
01 Introduction	Finite-joint PD(1) theorem; distinction between ordinary integers and prime predecessors.	Entire section read.
02 Preliminaries	Rough numbers, prime reciprocal sums, block sieve, long-prime polynomial interface.	Entire section read; external theorems and companion sieve input imported.
03 Determinant	One-sided Type II estimate; physical states, signed high moments, root residues and ideal line model.	Lines 1–440 and 593–648 read. Root-count proof and final error budget not fully inspected.
04 Memory	Primewise exact expansion, symmetric memory space, clean/dirty operators, global distinctness and pad removal.	Lines 1–436 and 968–1117 read. Clean/dirty and distinctness proofs 437–967 imported.
05 Major arcs	Long coefficients and sparse-frequency major-arc control.	Lines 1–43 read; remaining 422 lines uninspected.
06 Correlations	Two-sided marked estimate; endpoint comparison, local Mellin, divisor model and high-frequency bounds.	Lines 1–259 read; remaining 676 lines uninspected.
07 Extraction	Progressions, presieve, successful candidate groups, allocation, and removal of marks.	Entire 702-line section read; uses imported estimates above.
08 Poisson–Dirichlet	Factorial measures, size-biased triangular density, no-escape, sorting, passage to all primes.	Entire 254-line section read.
09 Appendix	Long-prime polynomial theorem via exponential sums, zero-free input and Mellin localization.	Lines 1–34 read; 624-line proof uninspected.

2.1 The arithmetic interface, stated without replacement

For fixed d and $\varepsilon > 0$, consider ordered tuples of distinct numerical primes $\ell_i \geq x^{\varepsilon}$ in fixed logarithmic intervals, whose product upper bound is at most $x^{1-\varepsilon}$. Put

$$f_x(u) = \sum_{\boldsymbol{\ell}} \mathbf{1}_{\ell_1 \cdots \ell_d | u}, \qquad C_x = \sum_{\boldsymbol{\ell}} \frac{1}{\ell_1 \cdots \ell_d}.$$

Both are bounded in the relevant dyad with constants depending on (d, ε). The source's extraction theorem asserts, for its fixed smooth dyadic weight Φ,

$$\sum_{p \text{ prime}} \Phi((p-1)/x)(f_x(p-1) - C_x) = o(x/L). \tag{1}$$

We import this statement where needed. It is not derived from Bernoulli independence or finite moment positivity. In particular, the prime-predecessor probability that a fixed prime divides $p-1$ is not literally $1/p$; the independent $1/p$ model below belongs to a specific divisor comparison, not to the prime sample itself.

The determinant input uses mark bands $\exp(L^{a_i}) < r \leq \exp(2L^{a_i})$, with $0.1 < a_1 < \cdots < a_K < 0.2$ fixed and $V_i = \sum_{r \in \mathcal{P}_i} 1/r \sim \log 2$. Roughness elsewhere uses $W = \exp(L^{0.24})$ and $V(W) = \prod_{r \leq W}(1 - 1/r)$. The two-sided correlation input has different candidate scales (small bands near exponents 0.27–0.36, large ones near 0.39–0.46) and must not be silently identified with the determinant bands.

The appendix's long-prime interface controls sums $\sum_{r \in I} \chi(r) r^{-1+it}$ for prescribed long prime intervals, small-modulus characters, and a large frequency window, with arbitrary fixed logarithmic savings. We have read its statement, not its analytic proof. Likewise, the companion positive block-sieve weights and summed progression remainders are imported hypotheses, not consequences of the coordinate changes in this report.

2.2 The determinant and high-moment normalization

The Cauchy square contains

$$mn - 1 = ah, \quad rs - 1 = bh, \qquad bmn - ars = b - a.$$

With pad product D, physical columns $P = (Dbm, s)$ and $Q = (Dar, n)$ satisfy $\det(P, Q)/D = bmn - ars$. This signed orientation is fixed throughout. The source takes $J = \lfloor L^{0.01} \rfloor$, $M = J + 1$, $R = \lceil L^{0.5}/2 \rceil$, and $N = 2R$. Ordered mark lists on a primitive physical state carry counting measure times $\prod_i V_i^{-M}$, not independent probability measure. A row copies J marks and creates one target mark per group; the joint row/adjoint normalization is $V_i^{-(M+1)}$.

For the compressed signed operator $A = GSTSG$ (where S is symmetrization and G goodness), the source controls

$$\frac{1}{UV} \sum_{P} \langle u_P, (AA^*)^R u_P \rangle \leq L^{-E_0 N}.$$

Here u_P is the indicator of *all* lists at P, without normalizing its norm. The compressed matrix $\langle u_P, (AA^*)^R u_Q \rangle$ is positive semidefinite in the ordinary coefficient space. Its largest eigenvalue is at most its trace even though the u_P are not unit vectors. Replacing them by unit vectors without changing coefficients would change the estimate. The spectral-Hölder and local-geometry deductions were read; the arithmetic signed-moment bound itself remains imported.

The source's replacement model gives a root real measure $\zeta(2)^{-1}\, du/U\, dv/V\, dr$ together with independent uniform projective lines over each relevant prime field. The root-residue count, geometry restrictions, and accumulated error are essential; the ideal line model is not an exact identity for physical roots.

3 Exact finite factorial coordinates

The private foundations used here are precise finite statements: an invertible evaluation/moment matrix changes coordinates on a specified finite span; covariance is computed from a common joint law; and an energy identity uses its actual Gram metric. None supplies arithmetic cancellation, realizability of arbitrary moment arrays, or an infinite-dimensional frame lower bound. These are the only transfers made from the private originals [5, 6, 7].

Proposition 3.1 (Finite Newton coordinates and Gram matrix)**.** *Let $\Omega = (\Omega_1, \dots, \Omega_g)$ have any probability law on $\prod_i \{0, \dots, n_i\}$. Write $(n)_r = n(n-1)\cdots(n-r+1)$, with $(n)_0 = 1$. Every function on this grid has a unique expansion*

$$F(n) = \sum_{0 \le r_i \le n_i} c_r \prod_i (n_i')_{r_i}, \quad n = (n_1', \dots, n_g'), \qquad c_r = \frac{(\Delta_1^{r_1} \cdots \Delta_g^{r_g} F)(0)}{\prod_i r_i!}.$$

For $m_r = \mathbb{E} \prod_i (\Omega_i)_{r_i}$ and $\phi_r = \prod_i (\Omega_i)_{r_i}$, its raw Gram matrix is

$$T_{rs} = \sum_{0 \le j_i \le \min(r_i, s_i)} \left[\prod_i \binom{r_i}{j_i} \binom{s_i}{j_i} j_i! \right] m_{r+s-j}. \tag{2}$$

The covariance matrix is $C = T - mm^{\mathsf{T}} \succeq 0$ and $\mathrm{Var}(\sum c_r \phi_r) = c^{\mathsf{T}} C c$. Degree a polynomials require joint moments of total degree at most $2a$.

Proof. The one-coordinate evaluation matrix $(n)_r$ is lower triangular with diagonal $r! \neq 0$; successive differences yield Newton's formula. Tensoring gives the grid formula. Two ordered injections of lengths r, s into n labels share j labels in $\binom{r}{j}\binom{s}{j} j!$ possible identifications. Counting their union proves

$$(n)_r (n)_s = \sum_j \binom{r}{j} \binom{s}{j} j! (n)_{r+s-j}.$$

Multiply this identity across coordinates and take expectations. Finally $c^{\mathsf{T}} C c = \mathbb{E}(F - \mathbb{E}F)^2 \ge 0$. The constant coordinate has zero covariance, so positive definiteness is neither required nor generally true. □

Joint, not marginal, factorial moments. In the source's candidate model, each prime has disjoint types "divides u" and "divides v". For a fixed number j of allocated factors their probabilities are

$$a_r = (r-1)^{j-1}/r^j, \qquad b_r = 1 - (1 - 1/r)^j.$$

Across different primes these categorical variables are independent. For a finite band $\mathcal{P}$, the exact joint probability generating function is

$$G(z, w) = \prod_{r \in \mathcal{P}} \{1 + a_r(z-1) + b_r(w-1)\}. \tag{3}$$

Consequently $\mathbb{E}(\Omega_u)_s (\Omega_v)_t = \partial_z^s \partial_w^t G(1,1)$. Equivalently it is the sum over ordered, mutually disjoint prime assignments of the corresponding products of a_r and b_r. A prime cannot occupy both types; replacing this by the product of the two marginal moments would discard a negative within-prime covariance. Formula (2) handles that dependence exactly.

A finite residual certificate. If a genuine common-law L^2 estimate gives $\|F - \widehat{F}\|_2 \le \epsilon$, then

$$|\mathbb{E}F - \mathbb{E}\widehat{F}| \le \epsilon, \qquad |\operatorname{Var} F - \operatorname{Var} \widehat{F}| \le 2\sqrt{\operatorname{Var} \widehat{F}}\,\epsilon + \epsilon^2.$$

Indeed, centering is an L^2 contraction and expand the square. For vector outputs the same proof gives the operator-norm covariance estimate with $\sqrt{\mathbb{E}\|\widehat{F} - \mathbb{E}\widehat{F}\|^2}$. The residual is not an independent noise term. These statements are useful for a rigorously truncated factorial bank, but a small probability of excluded configurations is insufficient unless its weighted second-moment contribution is bounded too.

4 Primewise memory: exact collapse and stability

Fix a path with visits $j = 0, \dots, N$, a prime r, and projective lines $\ell_j = [z_j]_r \in \mathbb{P}^1(\mathbb{F}_r)$. There are $m = r + 1$ possible lines. Let A be its active visits and let $q_j \in [0, 1]$; the source has $q_j = q$ internally and $\sqrt{q}$ at the two endpoints. Its exact primewise factor is

$$F_r = \frac{1}{m} \sum_{\ell} \prod_{j \in A} \mathbf{1}_{\ell = \ell_j} \prod_{j \notin A} (1 - (1 - q_j)\mathbf{1}_{\ell = \ell_j}). \tag{4}$$

This pays the line probability once per prime, not once per visit.

Proposition 4.1 (Line-class representation and perturbation)**.** *Let $C_\ell = \{j : \ell_j = \ell\}$. If A is empty, then*

$$F_r = 1 - \frac{1}{m} \sum_{\ell : C_\ell \neq \varnothing} \left(1 - \prod_{j \in C_\ell} q_j\right). \tag{5}$$

If $A \neq \varnothing$, incompatible active lines give zero; otherwise their common line ℓ_0 gives

$$F_r = \frac{1}{m} \prod_{j \in C_{\ell_0} \setminus A} q_j.$$

For fixed lines and active set, replacing the q_j by $\widetilde{q}_j \in [0, 1]$ changes F_r by at most $m^{-1} \sum_{j \notin A} |q_j - \widetilde{q}_j|$.

Proof. Condition in (4) on the selected line. Without active visits, its contribution is $\prod_{j \in C_\ell} q_j$, equal to 1 on an empty class. With active visits only a common line can contribute. For products of numbers in $[0, 1]$, telescoping gives $|\prod a_i - \prod b_i| \le \sum |a_i - b_i|$. Apply this to the disjoint classes or the unique active class. □

For numerical evaluation one may use the manifestly nonnegative formula $m^{-1}(m - c + \sum_{C_\ell \neq \varnothing} \prod_{j \in C_\ell} q_j)$, where c is the number of occupied lines. It takes linear work in the visits once the classes are available, without enumerating subsets of visits. This is a finite-path evaluation improvement, not a faster computation of the physical high moment. Across finitely many independent prime factors in $[0, 1]$, product telescoping bounds the absolute perturbation by the sum of these primewise errors. A relative error after dividing by a small baseline requires an additional lower bound.

4.1 Relation to the signed lifespan expansion

Put $\eta_j = 1 - q_j$ and $b'_r = 1 - \sum_j \eta_j/(r + 1)$. Expanding a class product in (5), and grouping a nonempty subset of size at least two by first and last visits, recovers exactly

$$b'_r + \frac{1}{r + 1} \sum_{a < c} \eta_a \eta_c \mathbf{1}_{\ell_a = \ell_c} \prod_{a < j < c} (1 - \eta_j \mathbf{1}_{\ell_j = \ell_a}).$$

For active visits, prefix/suffix telescoping supplies the optional signed endpoints. A prime resumed after inactivity must recover its previous line; sampling a new line would multiply by an extraneous $1/(r+1)$. The collapsed formula remains valid even if $b'_r \le 0$. By contrast the source's rescaled positive memory measure requires $b'_r > 0$. Thus positivity of this baseline is unnecessary for the finite identity, but remains necessary for that Hilbert-space implementation.

The source uses

$$\nu_i(r) = \frac{1}{V_i(r+1)b'_r}, \qquad \lambda_i(r,\ell) = \nu_i(r), \qquad \frac{\lambda_i^{\otimes k}}{k!}$$

on symmetric k-particle memory. The line coordinate has *counting measure*: there is no second factor $1/(r+1)$. The active list has ordered measure $\nu_i^{\otimes M}$. Factorials in the symmetric measure are essential to the adjoints: $\binom{k+a}{a}/(k+a)! = 1/(k!a!)$, and ordered promotion of a particles supplies the compensating $a!$. For fixed $\rho^2 > \sqrt{q}$, a surviving inactive hit contributes q_j/ρ^2 between the two endpoint factors ρ, hence exactly q_j. Global birth distinctness is a condition on a history, not a local one-step orthogonal projection.

The pad-removal identity is also exact:

$$\frac{1}{D}\prod_i V_i^{-(J+2)} = \left(\prod_{\text{all pad primes } r\in\mathcal{P}_i} \frac{1}{rV_i}\right)\prod_i V_i^{-2}.$$

It uses J ordered pads per group and inserts no $J!$. The probability of pad collisions alone does not bound a signed weighted square; the source multiplies the exponentially small collision estimate by absolute raw and comparison bounds.

4.2 A smaller sufficient cutoff at the truncation step

Proposition 4.2 (Local cutoff refinement)**.** *Suppose the source's absolute tilted-history estimate is available with constant C_K so that deleting histories with memory greater than B costs at most*

$$2^{-B+KN}\exp(C_K L^{0.74}), \qquad N = 2\lceil L^{0.5}/2\rceil. \tag{6}$$

For fixed $K, A > 0$ and $0 < \epsilon_0 \le 1$, this cost is at most $\epsilon_0 L^{-AN}$ whenever

$$B \ge KN + \frac{C_K L^{0.74} + AN\log L + \log(1/\epsilon_0)}{\log 2}.$$

In particular $B = \lceil L^\beta\rceil$ suffices eventually for every fixed $\beta > 0.74$, including $\beta = 3/4$, instead of the source choice $\lceil L^2\rceil$.

Proof. Take logarithms in (6) and rearrange. For fixed K, A, C_K, ϵ_0, both KN and $AN\log L$ are $o(L^\beta)$, as is $L^{0.74}$ when $\beta > 0.74$. □

The absolute estimate arises because at most KN stores occur, so a history crossing B has at least $B - KN$ extra births; doubling each birth produces the exponential moment used in Markov's inequality. This proof only refines that truncation step. We do not assert that all uninspected clean/dirty bounds have been reproved with the smaller space, nor infer a practical complexity bound for the full arithmetic argument. At $\beta = 0.74$ an unspecified coefficient C_K prevents the same conclusion.

5 Exact mark costs and removal of marks

For a finite prime band $\mathcal{P}_i$, let independent $I_r \sim \text{Bernoulli}(1/r)$ and $\Omega_i = \sum_{r\in\mathcal{P}_i} I_r$. Disjoint bands are independent *in this model*. For fixed $q \in (0,1)$ define

$$G_i(z) = \prod_{r\in\mathcal{P}_i} (1 + (z-1)/r), \quad H_i = \frac{\Omega_i q^{\Omega_i - 1}}{V_i}, \quad W = \prod_{i=1}^{K} H_i.$$

Define $H_i = 0$ when $\Omega_i = 0$. This W is the source mark $q^{\sum \Omega_i - K} \prod_i \Omega_i / V_i$.

Proposition 5.1 (Exact first and second mark moments)**.** *For nonempty bands,*

$$m_i := \mathbb{E}H_i = G_i'(q)/V_i, \tag{7}$$

$$s_i := \mathbb{E}H_i^2 = \{G_i'(q^2) + q^2 G_i''(q^2)\}/V_i^2, \tag{8}$$

$$m := \mathbb{E}W = \prod_i m_i, \qquad v := \text{Var}(W/m) = \prod_i \frac{s_i}{m_i^2} - 1. \tag{9}$$

These formulas include coincidences between represented prime slots and have no asymptotic remainder.

Proof. Differentiate $G(z) = \mathbb{E}z^{\Omega}$. For $z = q^2$, $\mathbb{E}\Omega^2 z^{\Omega-1} = G'(z) + zG''(z)$, using $\Omega^2 = (\Omega)_2 + \Omega$. Independence across the bands gives the two product identities. The mean is strictly positive. □

The second moment must use q^2, not q. The diagonal term $G_i'(q^2)$ is not optional: it accounts for repeated selection of the same represented prime in the two copies of a mark. This is exactly the distinction between ordinary powers of counts and falling factorials in the progression calculation.

Corollary 5.2 (Fixed-band Poisson cost and local tuning)**.** *If* $\max_{r\in\mathcal{P}_i} 1/r \to 0$ *and* $V_i \to \lambda > 0$*, then for fixed* K*,*

$$v + 1 \longrightarrow R_\lambda(q)^K, \qquad R_\lambda(q) = (q^2 + 1/\lambda)\exp(\lambda(q-1)^2).$$

For $\lambda = \log 2$*,* R_λ *has a unique minimizer in* $(0,1)$*, the root of*

$$\lambda q^3 - \lambda q^2 + 2q - 1 = 0. \tag{10}$$

This root is in $(0.54, 0.56)$*.*

Proof. Uniformly on a fixed bounded disk, $\log G_i(z) = (z-1)V_i + O(\sum r^{-2})$ and the error tends to zero. Thus G_i and its first two derivatives converge to those of $e^{\lambda(z-1)}$. Substitution into (7)–(9) gives the formula. Moreover

$$\frac{d}{dq}\log R_\lambda(q) = 2\lambda\left(q - 1 + \frac{q}{1+\lambda q^2}\right).$$

The bracket has derivative $1 + (1-\lambda q^2)/(1+\lambda q^2)^2 > 0$ on $[0,1]$ when $\lambda = \log 2 < 1$. It changes sign from negative to positive. Multiplication by $1 + \lambda q^2$ yields (10); direct endpoint inequalities give the stated bracket. □

This tunes variance for a fixed ideal array, not the whole proof: the required number K of bands in the arithmetic determinant estimate can itself depend on q. Even optimized, the normalized variance generally grows exponentially in K. A statement such as "bounded for fixed K" conceals that cost but does not remove it.

5.1 Covariance-weighted deweighting with an explicit bridge

Let Q be a finite positive arithmetic measure, not necessarily a probability measure. In the source one may take

$$Q_x(F) = \frac{L}{x} \sum_p \Phi((p-1)/x) F(p-1), \qquad \Phi \geq 0.$$

For model-normalized marks $X_a = W_a/m_a$, let C denote their covariance under a specified model law and choose $\lambda_a \geq 0$, $\sum_{a=1}^{b} \lambda_a = 1$. Set $Z = \sum_a \lambda_a X_a$.

Theorem 5.3 (Conditional finite deweighting bound)**.** *Suppose $Q(f^2) \leq H$, $|Q(X_a f)| \leq \eta_a$, and an* arithmetic *comparison has established*

$$Q((Z-1)^2) \leq C_{\mathrm{sieve}} \lambda^{\mathsf{T}} \mathsf{C} \lambda + \rho_0, \qquad \rho_0 \geq 0. \tag{11}$$

Then

$$|Q(f)| \leq \sum_a \lambda_a \eta_a + \sqrt{H} \sqrt{C_{\mathrm{sieve}} \lambda^{\mathsf{T}} \mathsf{C} \lambda + \rho_0}. \tag{12}$$

For independent model arrays with positive variances v_a, the variance-minimizing weights are $\lambda_a = v_a^{-1} / \sum_c v_c^{-1}$, with minimum $(\sum_a v_a^{-1})^{-1}$. Identical arrays give v/b.

Proof. Write $Q(f) = Q(Zf) + Q((1-Z)f)$ and apply Cauchy–Schwarz for the positive measure Q. This yields (12). In the independent case the quadratic form is $\sum_a v_a \lambda_a^2$; Cauchy–Schwarz applied to $\sum_a \lambda_a$ proves the inverse-variance minimum and its equality condition. If an array has zero variance, placing all weight there gives zero model variance but still requires (11). □

The theorem weakens *model independence* to a known covariance budget; it does not weaken the need for arithmetic comparison. For example a validated bound $\|\mathsf{C} - \widetilde{\mathsf{C}}\|_{\mathrm{op}} \leq \epsilon$ adds at most $\epsilon \|\lambda\|_2^2$ to the quadratic budget. The exact moments in Proposition 3.1 can compute a finite common-law covariance. Merely checking that a proposed matrix is positive semidefinite neither realizes that law nor proves (11) on primes.

5.2 Where the source earns the arithmetic bridge

The extraction section first proves progression comparisons for one or two mark arrays, including all coincident represented slots. Bonferroni truncation has depth of order $L^{0.01}$ and divisor size $\exp(O(L^{0.22}))$. The forced-absence cost for primes dividing a progression modulus is exponentially small at the mark scale; this is not simply independence in a residue class. The presieve then fixes an even h, sets $\gamma = c_0/(4h+3)$, and uses level $z^{4h+2} < x^{c_0}$ for $z = x^{\gamma}$. The surviving relative error is of order e^{-h}/γ with its stated uniformity.

Among $n \asymp \log L$ candidate groups, the success indicators satisfy $\mathbb{E} I_g \geq \beta > 0$ and uniformly $|\operatorname{Cov}(I_g, I_{g'})| \leq \epsilon_x \to 0$. Therefore

$$\mathbb{P}\left(\sum I_g < \beta n/2\right) \leq \frac{4}{\beta^2 n} + \frac{4\epsilon_x}{\beta^2}.$$

No rate $\epsilon_x = o(1/\log L)$ is needed. The failure mass is discarded while the probability representation is positive, *before* expanding its signed terms. Multiplying an earlier unquantified $o(x/L)$ by the later 3^n expansion would be invalid. Stripping all active candidate parts makes the coefficients allocation-invariant, and the exact identity

$$j^{\omega}(q_0/j)^{\omega - s} = j^s q_0^{\omega - s}$$

restores the allocation count.

Finally the source averages b disjoint arrays. Its upper-sieve comparison for the square yields, at fixed parameters, $\limsup Q_x((Z-1)^2) \ll C_K/b$. Combining this with the marked estimate gives the schematic bound $Ce^{-h}/\gamma + C\sqrt{C_K/b}$. The order is essential: fix $d, \varepsilon, c_0, h, \gamma, j$, then sufficient K, then finite b and the arrays; let $x \to \infty$, then $b \to \infty$, then $h \to \infty$ with new parameters as necessary. Neither optimal weights nor exact v authorize $b = b(x)$ or $K = K(x)$ without new uniform arithmetic remainders.

6 From interior factorial measures to transport

This section supplies complete probability proofs conditional on the source interior interface. It also specifies which useful quantitative bounds survive without an arithmetic rate.

6.1 The exact size-biased change of measure

For a deterministic mass-one sequence v, regard equal masses as distinct *labelled* balls. Choose successively without replacement, each time with probability proportional to remaining mass, and pad with zeros after exhaustion. Let $T_{x,1:d} = (T_{x,1}, \dots, T_{x,d})$ be the resulting size-biased prefix for V_x, and let $\nu_{x,d}$ be its law on

$$K_d = \{t \in [0,1]^d : \sum_j t_j \le 1\}, \qquad D_d = \{t_j > 0 : \sum_j t_j < 1\}.$$

Let $\mu_{x,d}$ be the factorial measure averaging the sum over ordered, distinct labelled balls of V_x. The source proves

$$\int \phi \, d\mu_{x,d} \longrightarrow \int_{D_d} \phi(t) \prod_{j=1}^{d} \frac{dt_j}{t_j} \quad (\phi \in C_c(D_d)), \tag{13}$$

for each fixed d. Our subsequent conclusions assume (13).

Numerical primes versus labelled factors. The arithmetic input (1) uses distinct numerical primes, whereas this measure distinguishes occurrences of the same prime. On a compact interior box $t_i \ge a > 0$, a repeated prime is large; the exceptional square-divisibility count is $O(x^{1-a/2})$ at the conservative dyadic threshold. The number of selected tuples is bounded by a constant depending on (a, d). This permits the source's replacement of the two conventions on compact interior supports, not at the small-prime boundary. The normalization $\log(p-1)$ differs from $\log x$ by a relative $O(1/L)$ on a fixed dyad. Smooth strips and compact mass bounds complete that step. We do not replace it by a claim of uniformity as $a \downarrow 0$ or $d \to \infty$.

The sequential sampling rule gives the exact identity on D_d

$$d\nu_{x,d} = w_d \, d\mu_{x,d}, \qquad w_d(t) = \prod_{j=1}^{d} \frac{t_j}{1 - \sum_{i<j} t_i}. \tag{14}$$

Thus its limiting interior density is

$$h_d(t) = \prod_{j=1}^{d} (1 - \sum_{i<j} t_i)^{-1}.$$

Put $A_j = t_j/(1 - \sum_{i<j} t_i)$, so $t_j = A_j \prod_{i<j}(1 - A_i)$. This triangular map from $(0,1)^d$ onto D_d has Jacobian $\prod_j \prod_{i<j}(1 - A_i)$, which cancels h_d. Hence $h_d(t)dt$ is the law ν_d of $(B_1, \dots, B_d)$ and has total mass exactly 1.

Lemma 6.1 (No-escape with a usable seminorm)**.** *Take any $\chi \in C_c(D_d)$ with $0 \le \chi \le 1$, and set*

$$\eta = 1 - \int \chi \, d\nu_d, \qquad e_0 = \left| \int \chi \, d(\nu_{x,d} - \nu_d) \right|,$$
$$e_1 = \sup_{\substack{F:K_d \to \mathbb{R},\ \mathrm{Lip}_{\ell^1}(F) \le 1 \\ F(0)=0}} \left| \int \chi F \, d(\nu_{x,d} - \nu_d) \right|.$$

For $w_{x,d} = W_1^{\ell^1}(\nu_{x,d}, \nu_d)$,

$$w_{x,d} \le e_0 + e_1 + 2\eta. \tag{15}$$

At fixed d, χ, (13) implies $e_0, e_1 \to 0$. One can choose such a cutoff with $\eta \le 2d\delta$ for any $0 < \delta < 1/2$, supported where all A_j lie in $[\delta/2, 1 - \delta/2]$.

Proof. The missing χ-mass for $\nu_{x,d}$ is at most $\eta + e_0$. Every admissible F satisfies $|F(t)| \le \|t\|_1 \le 1$. Split its integral difference into χF and $(1 - \chi)F$ to obtain (15) by the Kantorovich dual formula on the compact metric space K_d. Fixed tests converge by (14). The anchored 1-Lipschitz family is compact in uniform norm by Arzelà–Ascoli. A finite uniform net and total mass at most 2 show uniform convergence for the χF integrals. Finally take a product cutoff in the A coordinates, equal to 1 when every $A_j \in [\delta, 1 - \delta]$. Its missing uniform-cube mass is at most $2d\delta$. Its support stays compactly inside D_d: all t_j and the final residual are at least $(\delta/2)^d$ there. □

The seminorm e_1 involves only compact interior tests $\chi F w_d$ against $\mu_{x,d}$. It makes the input needed for a rate explicit. Pointwise convergence of individual test integrals is enough qualitatively at fixed d, χ, but no effective net error follows from an unspecified $o(1)$.

6.2 Exact missing mass and a finite-prefix transport bound

Lemma 6.2 (Sorted deletion and contraction)**.** *If a nonnegative summable sequence v has a selected subcollection s, padded with zeros, then $\mathrm{sort}(v)_i \ge \mathrm{sort}(s)_i$ for every i and*

$$\|\mathrm{sort}(v) - \mathrm{sort}(s)\|_1 = \sum v_i - \sum s_i. \tag{16}$$

Sorting is ℓ^1-nonexpansive for finite nonnegative vectors of the same length, and hence for infinite summable sequences.

Proof. The ith largest selected value cannot exceed the ith largest full value. Summing their nonnegative differences proves (16). For $a \ge b$ and $c \ge d$, $|a - c| + |b - d| \le |a - d| + |b - c|$; uncrossing a matching proves that matching the two sorted lists minimizes the ℓ^1 cost. This is at most the original matching cost. Truncate infinite lists and use (16) to pass to the limit. □

Theorem 6.3 (Explicit ranked transport budget)**.** *Let $R_{x,d} = 1 - \sum_{j=1}^{d} T_{x,j}$. For every x, d,*

$$W_1^{\ell^1}(\mathcal{L}(V_x), \mathrm{PD}(1)) \le \mathbb{E}R_{x,d} + w_{x,d} + 2^{-d} \tag{17}$$
$$\le 2w_{x,d} + 2^{1-d} \tag{18}$$
$$\le 2e_0 + 2e_1 + 4\eta + 2^{1-d}. \tag{19}$$

Assuming (13) for every fixed d, the left side tends to zero.

Proof. Couple V_x with its size-biased prefix sorted and zero-padded. Lemma 6.2 gives cost exactly $R_{x,d}$. Couple the two prefix laws optimally; sorting cannot increase their cost. Couple the GEM prefix to its full ranked sequence using the same uniform variables. Its missing mass is $\prod_{j\le d} U_j$, with expectation 2^{-d}. The triangle inequality for Wasserstein distance proves (17). The residual function $1-\sum t_j$ is 1-Lipschitz, so $\mathbb{E}R_{x,d} \le 2^{-d} + w_{x,d}$, proving (18). Use Lemma 6.1 for (19). At fixed d, first use a fixed cutoff and let $x \to \infty$; then improve the cutoff to obtain $w_{x,d} \to 0$. Finally let $d \to \infty$ in (18). □

Attribution and strength of the conclusion. This is an explicit cost refinement, not a new qualitative prime-factor law. For nonnegative mass-one sequences, coordinatewise convergence to a mass-one limit already implies ℓ^1 convergence: choose a finite prefix carrying all but ϵ of the limit mass, use finite-coordinate convergence there, and bound both remaining tails by mass conservation. A standard coupling formulation gives the corresponding probability deduction; bounded diameter then gives W_1 convergence. Thus the qualitative conclusion is essentially available from the source's finite-joint statement with its mass-one limit. The additional useful output is the explicit prefix/residual budget and the localized error seminorm, not a new topology theorem.

For the actual source, one first has dyadic averages with weights. Its final section passes to uniform primes up to real X by fixing a number J_0 of dyads, letting $X \to \infty$, and then $J_0 \to \infty$. The omitted prime proportion is $2^{-J_0} + o(1)$ by the prime number theorem. The same passage applies to bounded Lipschitz tests, whose anchored absolute value is uniformly bounded. Neither this observation nor (19) provides uniform estimates for $d = d(X)$ or $\delta = \delta(X)$ without additional arithmetic control.

7 Observables, limitations, and research directions

Corollary 7.1 (Controlled power sums)**.** *For $a \ge 1$, the map $F_a(v) = \sum_i v_i^a$ is a-Lipschitz on $\nabla_{\le 1}$. Under (13),*

$$\mathbb{E}\sum_i V_{x,i}^a \longrightarrow \frac{1}{a}, \qquad \left|\mathbb{E}F_a(V_x) - \frac{1}{a}\right| \le a\{\mathbb{E}R_{x,d} + w_{x,d} + 2^{-d}\}.$$

Proof. For $s,t \in [0,1]$, $|s^a - t^a| \le a|s-t|$; sum and apply (17). For GEM, $\mathbb{E}B_j^a = (a+1)^{-j}$, so monotone convergence and a geometric sum yield $\mathbb{E}\sum_j B_j^a = 1/a$. Sorting preserves the sum. □

For $0 < a < 1$ the corresponding inference fails. To see it even near the desired limit, let $P \sim \mathrm{PD}(1)$, replace its masses by $(1-\epsilon)P_i$, adjoin m equal masses ϵ/m, and sort. Call the result $P^{\epsilon,m}$. Coupling before sorting and using contraction gives $\mathbb{E}\|P^{\epsilon,m} - P\|_1 \le 2\epsilon$. Yet the new small masses contribute $\epsilon^a m^{1-a}$ to the a-power sum. Taking m sufficiently large makes this diverge while $\epsilon \downarrow 0$. For entropy $H(v) = \sum_i -v_i \log v_i$ the added contribution is $\epsilon \log(m/\epsilon)$. The exact identity

$$H(P^{\epsilon,m}) = (1-\epsilon)H(P) - (1-\epsilon)\log(1-\epsilon) + \epsilon\log(m/\epsilon)$$

shows the same obstruction. In fact $\mathbb{E}H(P) = 1$: for GEM, nonnegative-term integration gives $\mathbb{E}[-B_j \log B_j] = j/2^{j+1}$, whose sum is 1. Choosing $\log m \asymp 1/\epsilon$ prevents entropy means from converging. This is a manufactured mass-partition example, not a claim about actual prime factors. Uniform integrability of the desired observable, or a suitable small-factor energy bound, is an additional necessary control.

Advantages. Exact factorial coordinates preserve repeated-slot terms and expose the correct joint covariance; line-class collapse avoids unnecessary signed cancellation in a finite path; the mark variance formula displays the dependence on K; the local cutoff bound quantifies one memory expense; and the

transport estimate separates finite-prefix error from missing mass. Each improvement has a checkable domain and a complete local proof.

Costs and nonclaims. The finite factorial bank can be very large and poorly conditioned. Model independence does not transfer to prime predecessors for free. Line-class evaluation does not give a physical root-count bound. A reduced local cutoff does not certify the omitted clean/dirty analysis. Exact variance cannot make the number of mark arrays height-dependent without uniform comparison errors. No effective arithmetic rate is obtained. The probability arguments are classical in nature, and no historical priority is claimed for their basic identities.

Feasible next questions. One useful next step is to prove the upper-sieve square comparison uniformly in a controlled family of covariance-weighted arrays; then (12) would give an explicit optimization problem with genuine arithmetic error terms. A second is to recheck the uninspected memory bounds uniformly for $B = \lceil L^{3/4} \rceil$, tracking every dependence on B. A third is an effective compact-interior factorial estimate in a finite Lipschitz net, which could be inserted directly into (19). Finally, entropy or sublinear powers require a separate weighted small-factor tail estimate. None of these future inputs is asserted to have been proved here.

8 Literature, evidence, and contribution assessment

The October 8, 2026 refresh used only public titles and public URLs. Arratia–Kochman–Miller [2] supply the established multi-intensity route to Poisson–Dirichlet limits. Bharadwaj–Rodgers [3], whose arXiv page records revision v4 on April 9, 2026, distinguish full level-one limits from restricted-support correlations at positive level of distribution. Ford [4], v4 dated January 19, 2026, develops a shifted-prime Kubilius model and total-variation comparison; its page notes a correction to Lemma 3.3. These references reinforce, rather than remove, the need to distinguish a model from an arithmetic comparison theorem. This refresh inspected public primary metadata/abstracts and the indicated public pages, not every proof in those papers. The public GitHub snapshot URL could not be fetched through the web tool; local source evidence, not a successful web fetch, underlies our manuscript reading.

The applicable private passages were inspected locally only. Their titles and contents were not sent to public search. The first original gives finite moment/evaluation reconstruction and positive polynomial Gram examples; the second gives joint-law covariance and L^2 residual control; the third gives exact finite energy identities and metric-sensitive frame coordinates. None is cited as an arithmetic proof of (1). Incidental passages read beyond the relevant lemma are included in the ledger but not represented as contributions to this report.

The owned proof-audit separates proved local propositions from imported estimates. The author-run exact-rational test script checks factorial products, joint categorical moments, line-class identities and perturbations, mark first/second moments, inverse-variance minimization, and sorting/deletion formulas. Tests are finite sanity checks, not proofs of asymptotic arithmetic. The build record distinguishes native compiler availability from the exported MiKTeX result. The final visual record covers every final PDF page; the freeze binds the report, owned evidence, and final build artifacts without freezing mutable parent files. Exact counts and hashes are in that manifest.

On the five-point contribution scale, C1 denotes no demonstrated increment, C2 a limited increment, C3 a notable local increment, C4 a substantial advance, and C5 a very high contribution. **C3** is justified here by the combined source-specific cost formulas, conditional covariance bridge, local cutoff refinement, and explicit transport seminorm bound. It is not justified by rebranding the qualitative PD(1) conclusion as new, by the number of pages, or by applicability L3. C4 or C5 would require substantially stronger

arithmetic or effective uniform input not established in this report.

References

[1] OpenAI, *The Poisson–Dirichlet law for prime predecessors*, September 24, 2026. Locally inspected build at pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; parent reports text identity at captured October 8 commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Public source snapshot. Section/line locators in this report refer to that build.

[2] R. Arratia, F. Kochman, V. S. Miller, *Extensions of Billingsley's theorem via multi-intensities*, arXiv:1401.1555v1, 2014.

[3] A. Bharadwaj, B. Rodgers, *Large prime factors of well-distributed sequences*, arXiv:2402.11884v4, April 9, 2026.

[4] K. Ford, *Poisson approximation of prime divisors of shifted primes*, arXiv:2408.03803v4, January 19, 2026; journal DOI 10.1093/imrn/rnaf079.

[5] M. Subbarao, private original `TR-rev-1-old-tr250.tex`, local proof passages 155–213 and 455–540. Exact route and SHA-256 in owned read ledger.

[6] M. Subbarao, private original `TR-core-pat-16.tex`, local proof passages 216–255 and 305–354. Exact route and SHA-256 in owned read ledger.

[7] M. Subbarao, private original `TR1674-revised-v2.tex`, local proof passages 168–218 and 266–361. Exact route and SHA-256 in owned read ledger.